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MILITARY SPECIFICATION

TRANSMITTER; REMOTE COMPASS, THIN WING TYPE ML-1 (UNSTABILIZED)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers certain design requirements and all performance requirements for procurement of one type of remote compass transmitter which is intended for use with remote indicating and gyro stabilized compass systems for installation and operation in Military aircraft.

1.2 Classification - The assembly covered by this specification shall be designated Type ML-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitations for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416 - Plating, Cadmium (Electrodeposited)

Military

MIL-P-116 - Preservation, Methods of
MIL-C-5015 - Connectors, Electric, "AN" Type
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400 - Electronic Equipment, Aircraft, General Specification for

FSC 6605

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MIL-I-6181 - Interference Control Requirements, Aircraft Equipment
MIL-S-7742 - Screw Threads, Standard, Aeronautical
MIL-C-7762 - Compasses; Installation of
MIL-C-7834 - Compass, Swinging, Aircraft
MIL-P-7936 - Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625 - Anodic Coating, for Aluminum and Aluminum Alloy
MIL-C-17858 - Compass System: Magnetic Slaved, Directional Gyro Stabilized, 4°/HR, MA-1
MIL-C-21013 - Compass, Directional Reference System: Magnetic Slaved, Directional Gyro, Roll Stabilized, 3°/Hr., Type MF-1
MIL-C-21605 - Compass, Magnetic, Remote Slaved Gyro, Type S-2
MIL-D-70327 - Drawings, Engineering and Associated Lists

STANDARDS

MS25396 - Transmitter, Remote Compass (Thin Wing)
MS33586 - Metals, Definition of Dissimilar

PUBLICATION

ANA Bulletin No. 143 - Specifications and Standards; Use of

2.2 Availability of Documents -

(1) When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Qualification - The transmitter furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein and has been listed on, or approved for listing on, the applicable qualified products list.

3.2 Associated Equipment - The compass transmitter shall operate with the following compass systems, however the compass systems

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are not to be supplied in accordance with the requirements of this specification.

Nomenclature	Designation	Applicable Specification
Compass System: Magnetic Slaved, Directional Gyro Stabilized	Type MA-1	MIL-C-17858
Compass, Magnetic, Remote Slaved Gyro	Type S-2	MIL-C-21605
Compass, Directional Reference System, Magnetic Slaved	Type MF-1	MIL-C-21013

3.3 Materials - Materials should conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical Materials - Non-critical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals - Metals shall be of the corrosion resisting type or suitably protected to resist corrosion in salt spray, or atmospheric conditions to which the unit may be subjected when in storage or in normal service life.

3.3.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Standard MS33586.

3.3.3 Non-Magnetic Materials - Non-magnetic materials shall be used for all parts of the unit except where magnetic materials are essential

3.3.4 Fungus-Proof Materials - Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent or suitably protected as approved by the Bureau of Naval Weapons.

3.3.5 Selection of Materials - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

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3.3.5.1 Standard Parts - Standard Parts (MS, AN or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part number. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts without alteration; and provided the corresponding standard part numbers are referenced in the parts list and on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

* 3.4 Design - The remote compass transmitter shall include devices such as shock and vibration mounts, junction box or cables, etc., as necessary to satisfy the performance requirements specified in Section 4. The compass transmitter shall be operable as a direction sensing device. The transmitter will be designed to accurately detect the direction of the lines of force of the earth's magnetic field, and shall be capable of transmitting this information electrically to a compass system. The outline dimensions of the transmitter shall conform to Standard MS25396. The design of the transmitter shall permit installation and swinging procedures in accordance with the applicable requirements of Specifications MIL-C-7762 and MIL-C-7834.

* 3.4.1 The remote compass transmitter shall provide a minimum of ± 6° of index error removal with top or bottom mounting. The transmitter when properly energized by the earth's magnetic field, shall be capable of providing an indicator signal transformer with a suitable directional voltage signal for the compass systems referenced in paragraph 3.2. The secondary coil inductance (leg to leg) when measured at 1,000 cps with a standard laboratory impedance bridge shall be 75 ± 10 millihenries and the direct current resistance shall be 30 ± 3 ohms. When the primary coil is excited at 23.5 volts and 400 cps, the secondary coil signal frequency shall be 800 cps. The impedance of the primary coil when excited with 23.5 volts and 400 cps shall have a magnitude between 400 and 700 ohms at a phase angle between 50° and 70°. The corresponding impedance of the secondary coil (leg to leg) shall have a magnitude of between 200 and 300 ohms at a phase angle between 65° and 88° at the 800 cps frequency.

3.4.2 Pendulous Element - The earth inductor shall be mounted on a pendulous element so that it tends to be horizontal when at rest. The pendulous elements shall have a minimum of 27° of freedom in all directions with respect to the case, and shall have a period of not less than 0.2 second, nor greater than 0.75 second.

* 3.4.3 Compensator - The compensator shall be assembled with sufficient friction to prevent any turning of the magnetic elements by vibration as specified in the tests High Temperature Operation, Voltage Frequency variation, vibration error and vibration endurance.

3.4.3.1 Compensator Adjustment - One complete revolution of each compensating screw shall effect one-half cycle of compensation.

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Another complete revolution in the same direction shall effect the other half cycle. Each screw and associated surrounding compensator plate shall be index marked. Matching of the index on the screw with the fixed index of the plate shall indicate a zero position of the compensation (neutral position).

*3.4.3.2 Compensator Performance - The performance of the compensator shall meet the requirements of the test for Dielectric Strength, Neutral Position, Compensator Range and Compensator Interference.

3.4.4 Electrical Characteristics -

3.4.4.1 Input - Normal excitation for the transmitter primary circuit shall be 23.5 ± 0.5 volts, single phase, 400 ± 40 cps, alternating current.

*3.4.4.2 Output - The 800 cps signal from the transmitter secondary circuit shall be transmitted over a 3-wire line so that it may be connected to the stator of the signal transformer. With the transmitter placed in a normal field strength and excited with 23.5 ± 0.5 volts, single phase, 400 ± 5 cps, alternating current and the signal transformer rotor tuned for the transmitter's maximum 800 cps signal, phase rotation and maximum signal voltage should be as follows:

Place the transmitter and signal transformer rotor on the zero degree heading (null position), and record the 800 cps signal voltages AB and AC legs of the transmitter. Then rotate the transmitter and the signal transformer rotor to the 30° heading (null position). The voltage across AB leg shall increase and across AC leg decrease. With the transmitter left on the 30° heading, rotate the signal transformer rotor to the 120° heading. The line to line voltage across AB leg shall not be less than (140) ± 25 millivolts

3.4.4.3 Electrical Connectors - The electrical connectors shall be in accordance with Specification MIL-C-5015.

*3.4.4.4 Electronic Parts - Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the procuring activity.

* 3.5 Performance - The transmitter shall satisfy the performance requirements as specified in Section 4 of this specification.

* 3.6 Interchangeability - All transmitters manufactured in accordance with this specification shall be directly and completely

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interchangeable with each other with respect to installation and performance. Changes in part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.7 Special Tools - The design of the compass transmitter shall be such as to accommodate to the greatest extent disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.8 Physical Requirements -

*3.8.1 Weight - The weight of the transmitter without compensator shall not exceed 1.2 pounds. The weight of the compensator only shall not exceed 0.1 pounds.

3.8.2 Finish - Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used. The paint shall not be nutrient of fungi.

3.8.3 Aluminum Alloy Parts - Where practical, aluminum alloy parts shall be covered with anodic film conforming to Specification MIL-A-8625.

3.8.4 Brass and Steel Parts - Where practical, brass and steel parts shall be cadmium plated in accordance with Specification QQ-P-416.

3.8.5 Screw Threads - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.8.6 Marking - The transmitter shall include a lubber line, and a scale sector of ± 10° and a "Fore" indicating mark. All markings shall be durable to withstand usage encountered in service.

*3.9 Identification of Product - A nameplate shall be securely attached to the exterior of the case and shall be legibly and durably marked with the following information:

Bureau of Naval Weapons - U. S. Navy
TRANSMITTER: REMOTE COMPASS, THIN WING, TYPE ML-1
(UNSTABILIZED)
MS 25396-1
MIL-T-19576 (Wep)
Stock Number
Contract (or Order) No.
Manufacturer's Part Number
Manufacturer's Serial Number
Manufacturer's Name (omit address)

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3.9.1 Part number - The manufacturer's part number marked in the space provided in the nameplate shall be identical with the manufacturer's production drawing number, including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.10 Workmanship- All details of workmanship shall be in accordance with high grade aircraft instrument manufacturing practice. Particular attention shall be given to neatness and thoroughness of assembling, and freedom of parts from burrs and sharp edges.

3.11 Installation -

3.11.1 Installation Requirements - The compass transmitter shall be designed to permit installation in accordance with the requirements of Specification MIL-C-7762.

3.11.2 Installation Instructions - The contractor shall furnish with each instrument, one printed copy of instructions, with illustrations and diagrams covering the installation of the instrument. Previous to printing, two copies shall be furnished to the Bureau of Naval Weapons for approval. The instructions shall be printed on 8-1/2 X 11 or 17 X 11 inch paper.

3.11.2.1 Envelope - An envelope furnished by the contractor, containing the installation instructions shall be packaged with each unit and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS"

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests - The inspection and testing of the compass transmitter shall be classified as follows:

(a) Qualification Tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Acceptance Tests: Acceptance tests are those tests accomplished on all transmitters manufactured and submitted for acceptance under contract.

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*4.2 Qualification Tests - The qualification tests of the compass transmitters shall consist of all of the tests of this specification. The tests shall be conducted in the following sequence:

Individual Tests
Representative Sample Tests
Low Temperature Exposure
High Temperature Exposure
Vibration Endurance
Humidity
Fungus Resistance
Salt Spray
Radio Interference
Shock

*4.2.1 Qualification Test Samples - Qualification test samples shall consist of three samples of the manufacturer's model or type number submitted for testing.

*4.2.2 Authorization - Authorization to submit compass transmitter for qualification testing shall be obtained from the Bureau of Naval Weapons, Navy Department, Washington 25, D.C.. After obtaining authorization, each qualification test sample shall be plainly identified by securely attached durable tags marked with the following information:

Sample for Qualification Test
Submitted by (NAME) (DATE) for qualification tests in accordance with the requirements of Specification MIL-T-19576 under authorization (Reference Bureau of Naval Weapons letter authorizing the tests)
Transmitter, Remote Compass, Type ML-1
MS25396-1
Manufacturer's Part Number
Name of Manufacturer

The samples and certified test results showing conformance with all the requirement of this specification shall be forwarded to the laboratory as specified in the letter of authorization.

4.2.3 Resubmission of Samples - In case of failure of submitted samples, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly demonstrated that changes have been made in the product which the Bureau of Naval Weapons considers sufficient to warrant additional tests.

*4.3 Acceptance Tests - The acceptance tests of the compass transmitter shall consist of the individual tests and the representative sample tests of this specification.

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4.3.1 General Provisions - The contractor shall submit all transmitters for acceptance tests. The contractor shall be responsible for accomplishing the required acceptance tests in a laboratory (owned or engaged by the contractor) acceptable to the procuring activity. All acceptance tests shall be conducted under the close supervision of the Government Inspector. The contractor shall furnish test reports, in duplicate, showing the quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of the material during the course of manufacture shall in no case be construed as a guaranty of acceptance of the finished product.

*4.3.2 Individual Tests - Each transmitter shall be subjected to and shall conform with the requirements of the following tests:

Examination of Product
Dielectric Strength
Scale Error
Scale Error (Weak Field)
Neutral Position
Compensator Range
Compensator Interference
Tilt Error
Signal Gradient
Signal Strength
400 and 1200 Cycle Signal Strength

*4.3.3 Representative Sample Tests - Each representative sample shall be subjected to the tests listed below in the order shown:

Scale Error (Specified Compensation and Tilt)
Low Temperature Operation
High Altitude-Low Temperature Operation
High Temperature Operation
Voltage Frequency Variation
Vibration Error
Vibration

*4.3.3.1 Sample Selection - A representative sample test shall consist of three transmitters selected at random by the Government Inspector from each lot submitted for acceptance under contract.

4.3.3.2 Lot Determination - A lot shall consist of not more than one hundred transmitters that have passed the Individual Tests. Lots shall in general consist of transmitters produced in sequence.

4.3.3.3 Lot Acceptance - A lot will be accepted when the sample transmitters have been subjected to, and meet the requirements of, the Representative Sample tests.

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Additional Tests - The representative sample tests to which any sample is subjected may be augmented by other tests of the specification that are deemed necessary by the procuring agency or the cognizant Government Inspector.

4.3.4 Rejection and Retest - When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lots meet all the requirements of the specification. If investigation indicates that the defects may exist in items previously accepted, full particulars concerning the defects, including recommendations for correction shall be furnished for the procuring activity's approval. Upon approval of the correction a test shall be included in the Individual Tests (Sec. 4.3.2) to check all equipment for this requirement until reasonable assurance can be obtained that the defect has been satisfactorily corrected. Any item which has been rejected may be reworked or replaced to correct the defects. Transmitters rejected after retest shall not be resubmitted without specific approval of the cognizant inspector.

4.4 Test Conditions -

4.4.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 25°C (77°F). When tests are made with atmospheric pressure and temperature substantially different from the above values, proper allowance shall be made for the difference from specified conditions.

4.4.2 Vibration Stand - Whenever a vibration stand is specified it shall consist of a device which will vibrate at any desired frequency between 500 and 3000 cycles per minute and subject the compass transmitter to vibration such that a point on the vibration stand shall describe in a plane inclined 45° to the horizontal plane a circle of the diameter specified herein.

*4.4.3 Attitude - Unless otherwise specified, the transmitter shall always be tested in its normal operating position.

4.4.4 Supply Voltage - Unless otherwise specified, all tests shall be conducted with 23.5 ± 0.5 volts, single phase, 400 plus or minus 5 cps alternating current applied to the transmitter.

4.4.5 Field Strength - Unless otherwise specified, the compass transmitter shall be located in a standard magnetic field that is undirectional and unvarying such as the earth's magnetic field having a horizontal field strength of 0.18 ± 0.02 oersted and a vertical field strength of 0.54 ± 0.02 oersted. Where a weak field is specified the horizontal components shall be 0.08 ± 0.01 oersted, vertical component 0.54 ± 0.02 oersted.

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*4.4.6 Standard Signal Transformer - All tests shall be conducted with a standard signal transformer with the following electrical characteristics. The signal transformer shall consist of a three phase, three wire stator surrounding a single phase, two wire rotor. The maximum input (line to line) impedance of the stator with the rotor circuit open shall be 150 ± 15 ohms, at a phase angle of 82° ± 8°, when measured at 1,000 cps with a standard laboratory instrument such as a General Radio Type 650-A impedance bridge or an approved equivalent. The rotor impedance with the stator open shall be 10,000 ± 1,000 ohms at a phase signal angle of 85° ± 2°. When the rotor is tuned for maximum 800 cps signal and 90 ± 2 millivolts and 800 ± 5 cps is applied between one stator lead and a bus connecting the other two stator leads. The maximum 800 cps signal from the rotor shall be 900 ± 90 millivolts. The rotor must be rotated until maximum voltage is obtained.

4.4.6.1 The signal transformer shall be free of inherent transmission error, so that errors in the compass direction signal caused by its use shall be kept to a minimum.

*4.4.6.2 The signal transformer shall have a suitable dial, mounted concentric with its rotor shaft and a suitable pointer on the shaft so that angular displacement from 0° to 360° of the rotor may be determined to within ± 0.05°.

*4.4.7 Turntable - All tests specified herein shall be conducted with the transmitter mounted on a turntable which shall consist of a horizontal platform rotatable at a uniform rate up to 30° per second about a vertical pivot secured to a fixed base which carries a scale accurately graduated from 0° to 360° in increments of 0.1°. Provisions shall be included that will assure the specified rates of turn and indicate all the angles through which the table can be rotated from a fixed reference point.

*4.4.8 Standard Null Meter - The standard null meter shall have a minimum input impedance of one megohm and be capable of detecting the null position of the rotor of the standard signal transformer to within ± 0.05° when the maximum line-to-line signal voltage of 90 ± 10 millivolts is applied to the stator.

4.4.9 Harmonic Analyzer - A frequency sensitive harmonic analyzer such as a General Radio Type 713-B or an approved equivalent shall be used when required.

4.4.10 Warm-up Time - A ten minute warm-up will be allowed before the conduction of any test unless otherwise stated.

4.4.11 Measurements - All measurements shall be made with instruments whose accuracy has been verified. If tests are conducted at the contractor's plant, the accuracy of the instruments and test equipment shall be verified periodically by the contractor to the satisfaction of the procuring activity.

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4.5 Test Methods -

*4.5.1 Examination of Product - The government inspector shall determine in accordance with Specification MIL-E-5400 and by any means or tests deemed necessary that each transmitter conforms with all the requirements of this specification not covered by specific tests.

4.5.2 Dielectric Strength Test - The transmitter shall be subjected to a high potential test of 375 volts, 60 cps applied between each receptacle pin and the base, for a period of five seconds with no breakdown of the insulation. The dielectric tested shall be set for one milliampere sensitivity.

4.5.3 Scale Error Test - The transmitter shall be mounted on a turntable in a normal position with standard field strength, and standard input voltage. The transmitter secondary circuit shall be connected to the stator or a standard signal transformer whose rotor is connected to a standard null meter, as specified in 4.4.8. The transmitter and signal transformer shall be set at zero or "North" heading (rotor or null position). The transmitter and signal transformer shall be turned together in heading increments of 30°. First in one direction then the opposite direction for a total of 390° in each direction and the points of maximum and minimum error determined. These errors shall be subtracted algebraically and divided by 2. The calculated result shall not exceed ± 0.5°.

4.5.4 Scale Error Test (Weak Field) - The transmitter shall be set up with the signal transformer as specified in the Scale Error Test and the transmitter shall be oriented to give zero index error. The horizontal field strength shall be decreased to 0.08 plus or minus 0.01 oersted. The shift in null positive position of the signal transformer rotor shall not exceed ± 0.75°.

*4.5.5 Neutral Position Test - The transmitter without compensator shall be set at the 0° heading specified in the Scale Error Test. The compensator, with both north-south and east-west screws set at neutral, shall then be placed in its normal position on the transmitter. The resulting shift of the null position of the signal transformer rotor shall not exceed ± 1°. This test shall be repeated with the transmitter set on the 90° and 270° headings.

*4.5.6 Compensator Range Test - The transmitter without compensator shall be set at zero heading as specified in the Scale Error Test. The compensator with both north-south and east-west screws set at neutral, shall be placed in its normal position on the transmitter. The north-south screw shall then be turned 180° degrees. The shift in null position of the rotor shall be between 8° and 11°, plus or minus. The same screw shall then be turned 360 degrees in the same direction. The resulting shift from neutral position, while turning, in null position of the rotor shall be between 8° and 11°. The test shall be repeated with the transmitter set on the 90° heading, using the east-west screw and leaving the north-south screw at neutral. The corresponding shift in null position shall be from 8° to 11° on either side of neutral position.

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4.5.7 Compensator Interference Test - The transmitter without compensator shall be set at the zero degree heading as specified in the Scale Error Test. The compensator shall then be placed in its normal position on the transmitter, with the north-south screw set at neutral and the east-west screw set at 180° from neutral. The resulting shift in null position of the rotor shall not exceed ± 1.0°. The test shall be repeated with the transmitter set on the 90° heading, with the east-west screw set at neutral and the north-south screw set 180° from neutral position.

4.5.8 Tilt Error Test - The transmitter shall be set up with the signal transformer as specified in the Scale Error Test on the zero degree heading. The transmitter shall then be tilted 15° in pitch (Fore End up). It shall be rotated smoothly about a vertical axis to the 90° heading in a period of 20 to 30 seconds. The error between the signal transformer rotor and the transmitter shall not exceed 1.5°. Similarly, the transmitter shall be rotated to and readings taken on the 180°, 270° and 360° headings. The same limits shall apply. No vibration shall be applied in this test.

*4.5.9 Signal Gradient Test - The transmitter shall be set up and connected as in the Scale Error Test. With the signal transformer held at the null position and the transmitter rotated in 1° steps to 5° each side of null. The 800 cps output of the transmitter, when measured by the specified harmonic analyzer shall not be less than 13 millivolts per degree.

*4.5.10 Signal Strength Test - The transmitter shall be mounted on a turntable in a normal position with a standard signal transformer. The transmitter and signal transformer shall be set at zero or "North" heading. Then the heading of synchronization null shall be determined by moving of the transmitter. The signal transformer heading shall be displaced from the transmitter heading and the maximum 800 cps voltage recorded. This voltage shall not be less than 800 millivolts.

4.5.11 400 and 1200 Cycle Signal Strength Test - The transmitter shall be set up with the signal transformer as specified in the Scale Error Test. The signal transformer shall then be rotated for a maximum of 400 cps signal which shall not exceed 40 millivolts. The signal transformer shall then be rotated for a maximum of 1200 cps signal which shall not exceed 40 millivolts.

*4.5.12 Scale Error Test (Specified Compensation and Tilt) - The transmitter without the compensator shall be set at the zero degree heading as specified in the Scale Error Test. The compensator with both the north-south and east-west screws set at neutral position, then shall be placed in its normal position on the transmitter. The north-south screw shall be rotated clockwise until 6° of compensation is set in. The transmitter shall be set at the 90° heading and the east-west screw shall be rotated clockwise until 6° of compensation is set in. The transmitter with a compensator is to be given a Scale Error Test. The transmitter with a compensator is to be tilted 15° fore end up, and then the Scale Error Test is to be repeated. The difference in the respective headings after the index errors have been removed from the two Scale Error Tests shall not exceed ± 2°.

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4.5.13 Low Temperature Operation Test - The transmitter shall be energized from the standard power supply and shall be subjected to a temperature of -55 + 2 C. (-67 ± 3.6 F.) for a period of 4 hours. The transmitter may be removed from the cold chamber and while at the specified temperature shall be subjected to and meet the requirements of the Scale Error Test except that the Scale Error Test tolerances shall not exceed ± 1.0 . The adjusting screws on the compensator shall operate without excess binding or freedom.

*4.5.14 High Altitude - Low Temperature Operation Test - The transmitter shall be energized from the standard power supply and shall be subjected to a pressure equivalent to 50,000 ± 500 feet at a temperature of -55 ± 2 C(-67 ± 3.6 F) for a period of 4 hours. The instrument may then be removed from the chamber, examined for oil leads, and while at the specified temperature, the instrument shall be subjected to and shall meet the requirements of the Scale Error Test except that the Scale Error Test tolerances shall not exceed ± 1.0° . The adjusting screws on the compensator shall operate without excess binding or freedom.

4.5.15 High Temperature Operation Test - The transmitter shall be energized from the standard power supply and shall be subjected to a temperature of 70°C. (158° F.) for a period of 4 hours. The transmitter may then be removed from the heat chamber and while at the specified temperature shall be subjected to and meet the requirements of the Scale Error Test except that the Scale Error Test tolerances shall not exceed ± 1.0°. The adjusting screws on the compensator shall operate without excess binding or freedom. After the transmitter has been saturated at room temperature for a minimum of four (4) hours, it shall be subjected to and meet the requirements of the Individual Test. There shall be no damage which would adversely affect subsequent operation. This test may be combined with the High Temperature Exposure test at the discretion of the testing agency.

4.5.16 Voltage-Frequency Variation Test - The transmitter shall be set up as specified in the Scale Error Test with signal transformer, on the zero degree heading. The supply voltage shall then be reduced to 21.5 volts and the frequency to 360 cps. The shift in null position of the signal transformer rotor shall not exceed ± 0.5°. The voltage shall then be increased to 24.5 volts and the frequency to 440 cps. The shift in null position from normal shall again not exceed ± 0.5° . This test shall be repeated on two other headings not less than 100° apart.

*4.5.17 Vibration Error Test - The transmitter shall be vibrated so that a point on its case describes a circle in a plane inclined at 45° to the horizontal. The transmitter shall be operated with the signal transformer aligned on the zero degree heading before vibration. The transmitter shall then be vibrated at frequencies of 600, 800, 1000, etcetera, cycles per minute through 2500 cycles per minute with a total amplitude of 0.003 to 0.005 inch. At each frequency, the shift in null position of the signal transformer rotor shall not exceed ± 1°from the normal heading. The plane of the circle of vibration shall contain the East-West axis. The test shall be repeated on two other headings not less than 100° apart. The compensator headings shall not change.

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4.5.18 Vibration Test - The transmitter shall be vibrated at a frequency of 900 cycles per minute with a total amplitude of 0.125 inch in a horizontal plane for a period of 100 hours. The transmitter shall be mounted on the Vibration Machine in such a manner that the zero index line on the transmitter mounting flange is 45° from the direction of motion. The transmitter shall then be subjected to and shall meet the requirements of the Scale Error Test and the Tilt Error Test.

4.5.19 Low Temperature Exposure Test - The transmitter shall be subjected to a temperature of -65 ± 2°C (-85 ± 3.6°F) for a period of 48 hours. The temperature shall then be raised to -55 ± 2°C (-67 ± 3.6°F) for a period of 24 hours. The transmitter while at the specified temperature shall be subjected to and shall meet the requirements of the Scale Error Temperature Compensation Test.

*4.5.20 High Temperature Exposure Test - The Transmitter shall be subjected to a temperature of 93° ± 2°C (200° ± 3.6°F) for a period of 24 hours. While still at this temperature, the transmitter shall be subjected to the "Individual Tests", except interference. After the chamber has been returned to room temperature and the transmitter has saturated at room temperature for a minimum period of 4 hours, it shall be subjected to and shall meet the requirements of the "Individual Tests". There shall be no damage which would adversely affect subsequent operation.

*4.5.21 Vibration Endurance Test - The requirements of this test shall constitute a method to detect mechanical and functional weaknesses of the transmitter. Compliance with this test shall be accomplished by mounting the transmitter on a vibration stand in its normal operating position and vibrating it in a vertical plane at a frequency of 55 cps and double amplitude of 0.064 inch for a period of two hours. The compensator and detector shall not be damaged nor shall the compensator setting be changed. At the end of this test, the transmitter shall be physically examined and there shall be no evidence of looseness in the mechanism as a result of this test. The transmitter shall then be subjected to and meet the requirements of the "Individual Tests".

4.5.22 Humidity Test - The transmitter shall be mounted in its normal operating position in a chamber and maintained at a temperature of 70 ± 2°C. (158 ± 3.6°F.) and a relative humidity of 95 ± 5% for a period of 6 hours. Distilled water shall be used to obtain the required humidity. At the end of the 6 hour period, the heat shall be shut off and the transmitter shall be allowed to cool for a period of 18 hours in this atmosphere, in which the humidity rises to 100 % as the temperature decreases. This cycle shall be repeated a sufficient number of times to extend the total time of test of 120 hours (5 cycles). At the end of the 120-hour (5 cycle) period and within one hour after the transmitter has been removed from the chamber, it shall be subjected to and meet the requirements of the "Individual Tests". Examination for corrosion and other deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the transmitter.

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*4.5.23 Fungus Resistance Test - The transmitter shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 14 days. At the option of the qualifying activity, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument.

4.5.24 Salt Spray Test - The transmitter, with external connections made to simulate installed conditions, shall be subjected to a salt spray in accordance with Procedure I (Par 4.6) of Specification MIL-E-5272c. At the end of the spray period, the transmitter shall be subjected to the Scale Error Test.

4.5.25 Radio Interference Test - The transmitter shall be operated at rated voltage and tested for conducted and radiated noise interference in accordance with Specification MIL-I-6181.

4.5.26 Shock Test - The transmitter shall be tested in accordance with Procedure V of Specification MIL-E-5272. After the testing procedure, the transmitter shall be subjected to and shall meet the requirements of the Scale Error Test.

5. PREPARATION FOR DELIVERY

5.1 Packaging -

*5.1.1 Level A - The transmitter shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method IId, using metal reusable containers (see 6.2)

5.1.2 Level C - When this level is required, packaging shall be in accordance with standard commercial practice.

5.2 Packing -

*5.2.1 Levels A and B - The instrument shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C - The instrument shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers' rules and regulations in effect at the time of shipment.

*5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

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5.3.1 Precautionary Marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE

DELICATE INSTRUMENTS

HANDLE WITH CARE

6. NOTES

6.1 Application - The transmitters covered by this specification are intended for use where a magnetic heading transmitter is required in conjuction with Compasses or All Attitude and Master Flight Gyro Reference Systems.

*6.2 Ordering Data - Requisitions, contracts and orders should state the quantity and MS part number of the transmitter desired and the levels of packaging (see 5.1) and packing (see 5.2) to be furnished.

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity.

6.4 Letter of Authorization - The letter of authorization is the letter from the qualifying agency to the manufacturer, authorizing the manufacturer to submit the qualification test samples to the Government for the Qualification Tests. The letter will contain authorization, shipping instructions and other relevant information.

6.5 Qualifying Agency - The Bureau of Naval Weapons is the qualifying agency for this specification. The Naval Air Development Center (Aeronautical Instruments Laboratory), Johnsville, Pennsylvania will perform the tests specified herein and forward the test results to the Bureau of Naval Weapons for approval. The Bureau of Naval Weapons will notify the manufacturers whether or not their products have been placed on the Qualified Products List.

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6.6 It is to be understood that the compass transmitters supplied under contract shall be identical in every respect to the sample tested and found satisfactory, except for changes previously approved by the Bureau of Naval Weapons. Any unapproved changes from the Qualification Test Sample shall constitute cause for rejection.

6.7 An asterisk preceding a paragraph number denotes a change in that paragraph from the basic specification.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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MILITARY SPECIFICATION

TRANSMITTER, REMOTE COMPASS, THIN WING TYPE ML-1 (UNSTABILIZED)

This amendment forms a part of Military Specification MIL-T-19576A(WEP) dated 1 February 1961 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment MIL-T-19576A(WEP) is changed to MIL-T-19576A(AS) and throughout the specification any reference to Bureau of Naval Weapons shall be changed to Naval Air Systems Command

Page 1, paragraph 1.1, line four after the word compass: Add "and attitude heading reference".

Page 1, paragraph 2.1., under Specifications Federal: Add

"QQ-N-286	Nickel-Copper-Aluminum"

Page 2, paragraph 2.1, under Specifications Military: Add

"MIL-A-23670	Attitude Heading Reference System; Type AN/ASN-50
MIL-C-26524	Calibration Set, Magnetic Compass, Type MC-1/-2
MIL-C-38418	Control Assembly, Gyroscope, Attitude; Type A/A24G-39
MIL-C-81412	Compass Set, Gyroscope Reference; Type AN/ASN-75"

Page 2, paragraph 2.1, under Standards Military: Add

"MIL-STD-810	Environmental Test Methods
EI-455	Installation Instructions for Type ML-1 Remote Compass Transmitter"

Page 2, paragraph 3.2: Delete first sentence and substitute

"The compass transmitter shall operate with the following compass and attitude heading reference systems, however the compass and attitude heading reference systems are not to be supplied in accordance with the requirements of this specification".

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Page 3, paragraph 3.2: Add

"Nomenclature	Designation	Applicable Specification
Attitude Heading Reference System	Type AN/ASN-50	MIL-A-23670
Control Assembly, Gyroscope Attitude	Type A/A24G-39	MIL-C-38418
Compass Set, Gyroscope, Reference	Type AN/ASN-75	MIL-C-81412

Page 4, paragraph 3.4: Add

"The transmitter shall be suitable for use with the MC-1 and MC-2 Magnetic Compass Calibrator Set, MIL-C-26524" (Sperry Part Nos. 2592080-4, 5, 7, 8).

Page 4, paragraph 3.4.1: Add

"The mounting flange and bowl of the transmitter shall be fabricated of material with a modulus of elasticity in excess of 22 x 10^6 PSI in accordance with QQ-N-286.

Page 5, paragraph 3.4.4.2, second paragraph final sentence: Change

"(140) ± 25 millivolts" to "to 125 millivolts minimum."

Page 6: Add

"3.8.7 Nickel, Copper and Aluminum Parts

Where required, parts and fabrication of materials shall conform to Specification QQ-N-286."

Page 7, paragraph 3.11.1: Add

"and EI-455."

Page 7, paragraphs 3.11.2 and 3.11.2.1: Delete both paragraphs.

Page 9, paragraph 4.3.2. Add "Sensitivity and Crosstalk Tests"

Page 9, paragraph 4.3.3: Add as the first required test: "Individual Tests"

Page 10, top of page: Add paragraph number "4.3.3.4"

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Page 10, paragraph 4.4.2: Add

"or a device which will vibrate at the requirements specified in MIL-STD-810."

Page 11. Add

"4.4.12 Vibration. Unless otherwise specified, the detector shall be subjected to vibration at a frequency from 25 to 35 Hz so that a point on the transmitter will describe, in a vertical plane, a circle with a diameter of not less than 0.002 nor more than 0.005 inch."

Page 14, paragraph 4.5.17, third sentence: Change

"cucles" to "cycles"

Page 15, paragraph 4.5.21: Delete and Substitute

"4.5.21 Vibration Endurance Test - The transmitter shall be operating and monitored and be vibrated in accordance with MIL-STD-810, Figure 514.1 Curve Z, Procedure I, Part 1. The transmitter shall then be subjected to and meet the requirements of the Scale Error Test."

Page 16, paragraph 4.5.26: Delete and substitute

"4.5.26 Shock Test - The transmitter shall be tested in accordance with MIL-STD-810, Figure 516.2, Procedure I. After the testing procedure, the transmitter shall be inspected to and shall meet the requirements of the Scale Error Test."

Page 16 Add

"4.5.27 Sensitivity and Crosstalk Test

The sensitivity and crosstalk tests shall be performed with the transmitter connected to the standard signal transformer as specified in the Scale Error Test (4.5.3).

4.5.27.1 Sensitivity

The transmitter shall set to the 0° or north heading and the signal transformer set to a heading of 90 ± 0.05° greater than the transmitter. A direct current shall be applied from Pins A to F of MS 25396 and varied until a null is indicated on the standard null meter. The direct current shall be 1.00 ± 0.05 milliamperes.

4.5.27.2 Crosstalk - 180°

The transmitter shall be set at the 0° heading and a direct current of

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2.00 ± 0.1 milliamperes shall be applied from pins A to F of MS 25396. This shall produce a 180° heading in the transmitter and the error at this heading shall not differ from the error obtained in the scale error test at the 180° heading by more than 0.8° when corrected for the misalignment of the transmitter and the readout.

4.5.27.3. Crosstalk - 270°

The transmitter shall be set at a 90° heading and a direct current of 1.15 ± 0.1 milliamperes from pins C to B of MS25396. This shall produce a 270° heading in the transmitter and the error at this heading shall not differ from the error obtained in the scale error test at a 270° heading by more than 0.8° when corrected for the misalignment of the transmitter and the readout."

Project Number 6605-N242

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